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Mining Insider Threats in Enterprise Systems Using Behavioral Data and Cyber Forensics

Dhruv Dhayal¹, Pratham Aggarwal², Manzoor Ansari³ (Corresponding Author), Masood Alam⁴, and, Murad Younas⁵

^{1,2,3} Department of Computer Science and IT,

Institute of Information Technology & Management, GGSIP, University New Delhi, India

^{4,5} Department of Mathematics and IT, Center for preparatory Studies, Sultan Qaboos University

¹dhayaldhruv271@gmail.com, ²aggarwalpratham2602@gmail.com, ³manzoor.ans@gmail.com,

⁴malam@squ.edu.om, ⁵m.younas@squ.edu.om

Abstract.

Insider threat detection faces challenges across the banking and stock exchange industries, with the digitization of financial infrastructure speeding up. Traditional security measures would increasingly lose their efficacy against sophisticated internal actors having legitimate access credentials. Presented here is an advanced ML detection framework, Threat Sense, that uniquely combines behavioral analytics, temporal pattern recognition, and digital forensics capabilities for enterprise financial systems protection. The framework applies multivariate temporal sequence mining algorithms to detect suspicious behavioral patterns with a maximum accuracy rate across varying transaction volumes and user activity patterns to know how the firms manage and handle by financial firms with secure transactions. Testing for its implementation across three very different financial institutions confirmed the best operation within normalized data environments, with detection power varying directly with the quality of data and precision of behavioral baseline measurements. Continuous monitoring is offered by the system; actionable security alerts are generated while forensic evidence chain-of-custody is scrupulously preserved for later use in legal proceedings. Unlike conventional detection systems, ThreatSense managed to identify attempts at unauthorized financial data access, subtle transaction manipulations, and highway-grade data exfiltrations in the course of penetration testing exercises. Future development areas will include more granular behavioral modeling techniques.

Keywords: Insider Threat Detection, Data Mining, Cyber Forensics, Behavioral Analytics, Financial Security, Anomaly Detection, Machine Learning, Temporal Sequence Mining, Enterprise Systems, Regulatory Compliance

1 Introduction

Financial institutions today operate in an increasingly digitized environment where vast amounts of sensitive data and transactions are processed continuously. Banks, stock exchanges, and financial firms manage critical operations through enterprise systems that handle everything from customer records to high-value transactions worth billions of dollars daily. While these institutions have traditionally focused their security efforts on external threats, insider threats pose a particularly insidious challenge that conventional security measures often fail to address effectively. Insider threats—malicious activities perpetrated by individuals with legitimate access to an organization's systems—represent a significant vulnerability in the financial sector [1],[2]. According to recent industry reports, approximately 34% of data breaches involve internal actors, with financial services experiencing 35% higher costs from insider threats compared to other industries [5]. The privileged access these individuals possess enables them to bypass traditional security controls, making detection particularly challenging [3],[4]. Moreover, these threats often remain undetected for extended periods, averaging 287 days before discovery, allowing malicious actors to cause substantial financial damage and erode customer trust in affected institutions. The financial impact extends beyond direct monetary losses to include regulatory penalties, legal liabilities, and significant reputational damage. The intersection of data mining and cybersecurity presents a promising approach to addressing this critical gap in financial system security. Data mining techniques excel at extracting meaningful patterns from vast quantities of data, while cybersecurity forensics provides methodologies to preserve evidence and establish attribution. By combining these disciplines, our research introduces a novel framework that leverages the strengths of both fields to identify potential insider threats before they result in significant damage [6]. This integrated approach is particularly valuable for financial

institutions processing high-volume transaction data, where traditional rule-based detection systems prove inadequate for identifying subtle behavioural anomalies that may indicate malicious insider activity.

In recent years, financial institutions have increasingly become reliant on highly digitized infrastructures to support their daily operations. As transactions and data exchanges occur at unprecedented speed and scale, the risk of internal threats—particularly from employees or trusted partners—has risen substantially. Unlike external cyberattacks, insider threats are far more difficult to detect as they often blend seamlessly with legitimate operations, exploiting trusted access to commit data breaches, fraud, and sabotage. This growing problem is amplified by the complexity of modern enterprise systems that handle millions of transactions daily. Traditional perimeter-based security approaches are no longer sufficient in environments where internal actors can bypass firewalls, evade basic anomaly detection, and manipulate data without triggering conventional alarms. The financial consequences of such undetected activities can be catastrophic, including monetary loss, legal penalties, and damage to institutional reputation. Moreover, regulatory frameworks such as GDPR, HIPAA, and ISO/IEC 27001 impose strict data protection and compliance requirements. Enhancing Financial Security through Data Mining and Cybersecurity: An Integrated Approach for Threat Detection and Prevention

Financial organizations produce huge amounts of data throughout the international banking industry. Our study uses sophisticated data mining methodologies to efficiently manage this huge influx of data while obtaining usable security intelligence [7],[8]. Our methodology starts from strong data ingestion and preprocessing, whereby disparate financial data sources are cleaned, standardized, and timestamped, minimizing data volume yet improving quality for examination [11],[12]. By advanced feature engineering, we build behavioral fingerprints, temporal profiles, and contextual augmentation that reflect rich user interactions with the financial system [9],[12],[14]. Dimensionality reduction algorithms extract the most discriminating features, simplifying the feature space without sacrificing detection performance [11]. The analytical foundation of our approach uses several complementary methods—supervised classification for familiar threat patterns, unsupervised anomaly detection for new threats, and sequential pattern mining for compound attack sequences—operating synergistically by weighted ensemble techniques to provide high detection rates with controllable false positive rates. Supporting these data mining functions, our architecture incorporates essential cybersecurity elements to provide transaction safety and proof protection. Ledgering inspired by blockchain gives unalterable transaction history that safeguards against privileged insider manipulation, while digital signatures facilitate non-repudiation for financial transactions [13]. The security framework applies segregation of duties and out-of-band logging to avoid monitoring bypass attempts by insider threats. For threats that are discovered, automated forensic practices save digital evidence with adequate chain-of-custody, supporting effective investigations while ensuring evidence admissibility for future legal cases.

1.1 Problems Definition

Insider threats occur when insiders such as employees, contractors, or partners misuse authorized access to violate the confidentiality, integrity, or availability of enterprise information and services [1],[2],[5]. Normal security controls of the traditional perimeter type, as well as rule-based monitoring, cannot detect these threats on time since malicious activities mostly look like routine user behavior [3],[11]. The research issue is to create a data-oriented framework that extracts rich behavioral logs and employs cyber-forensic methods to identify, attribute, and reconstruct insider attacks in large-scale enterprise settings—preferably in (near) real time with evidentiary rigor [4],[6],[13]. Traditional security systems primarily focus on perimeter defence—such as firewalls, intrusion prevention systems, and static rule-based monitoring—which are ineffective against sophisticated insider tactics. These individuals often exploit their familiarity with internal processes, tools, and blind spots in monitoring to carry out malicious activities without raising immediate suspicion. Additionally, most insider attacks occur over long durations in small increments, blending malicious behaviour with routine actions. Complicating the problem further is the lack of real-time analysis and attribution.

1.2 Background and Related Work

To address insider threats in enterprise systems, numerous studies have applied data mining and cybersecurity techniques. The following table summarizes key findings, contributions, and how each work aids in solving the insider threat problem using behavioral analysis and forensic intelligence.

Table 1: Comparative Analysis of Existing Insider Threat Detection Approaches Leveraging Data Mining and Cyber security

Author	Findings	Contribution	Problem Solution
Koli et al. (2025) [9]	Real-time insider risk detection using autoencoders	Scalable AI-based detection model	Uses anomaly detection on behavioral data for secure enterprise monitoring
Lopez & Sartipi (2020) [10]	Insider threat detection using LSTM on user logs	Sequential modeling of user behavior	Applies data mining to detect suspicious user patterns over time
Zhang et al. (2023) [14]	Deep clustering on multi-source behavior logs	Captures complex, cross-source user behavior	Enhances data fusion to spot subtle insider threats in large systems
Yuan & Wu (2020) [11]	Survey of deep learning challenges for insider threats	Identified data, model, and deployment issues	Guides secure system design combining AI and forensic behavior mining
Li et al. (2023) [12]	User behavior modeled with graph neural networks	Relationship-aware insider detection	Leverages graph mining for user intent analysis and forensic tracing
Greitzer et al. (2016) [13]	Anomaly detection is superior to rule-based systems	Meta-analysis of detection models	Promotes behavior-based cybersecurity over static rule engines

2 Proposed Methodology

The suggested approach offers an inter-disciplinary solution based on data mining, cyber security, and cyber forensic methods to preventively identify and neutralize internal threats inside corporate financial networks such as bank institutions and stock exchange systems. The approach starts by gathering colossal quantities of behavior data from various sources such as user activity traces, transaction records, access control systems, and communication records. This data is preprocessed by cleaning, normalizing, and anonymizing to obtain both user anonymity and data quality. After fine-tuning, pertinent behavioral features are derived—e.g., login history, transaction history, file access patterns, and command-line patterns—to build normal user behavior models for the organization [12],[13]. These behavior profiles are used as baselines on which anomalies are tracked. Advanced data mining methods are used to identify potential insider threats. Clustering algorithms are employed for finding user behavior outliers, and classification algorithms are utilized for tagging known patterns of threats. Deep learning algorithms like LSTM and autoencoders also detect sequential behavioral anomalies and make the detection algorithm resilient even against subtle or emerging threat vectors [9],[10]. For improving the context awareness of the system and eliminating false positives, hybrid recommendation models with both collaborative and content-based filtering are also employed [11]. In contrast, effective cybersecurity solutions—role-based access control, secure authentication mechanisms, and real-time monitoring systems—are deployed to protect confidential transactions and data transfers [13]. Any identified anomaly will automatically generate alarms and auto-response actions like session teardown or access revocation [9]. At the same time, cyber forensic tools are utilized to track the origin, motivation, and possible harm caused by the insider activity. Graph-based models, particularly graph convolutional networks, are utilized to visualize and analyse user relationships and system interactions and hence contribute to deeper forensic analysis. The entire architecture is deployed as a scalable and modular prototype system for effortless integration with existing enterprise security infrastructure. Its efficiency is rigorously tested using real or simulated insider threat attacks, and detection precision, false positive rate, latency, and scalability are the most vital metrics. By this approach, not only are financial institutions more secure from internal threats but incident response can also be effortlessly and intelligently made.

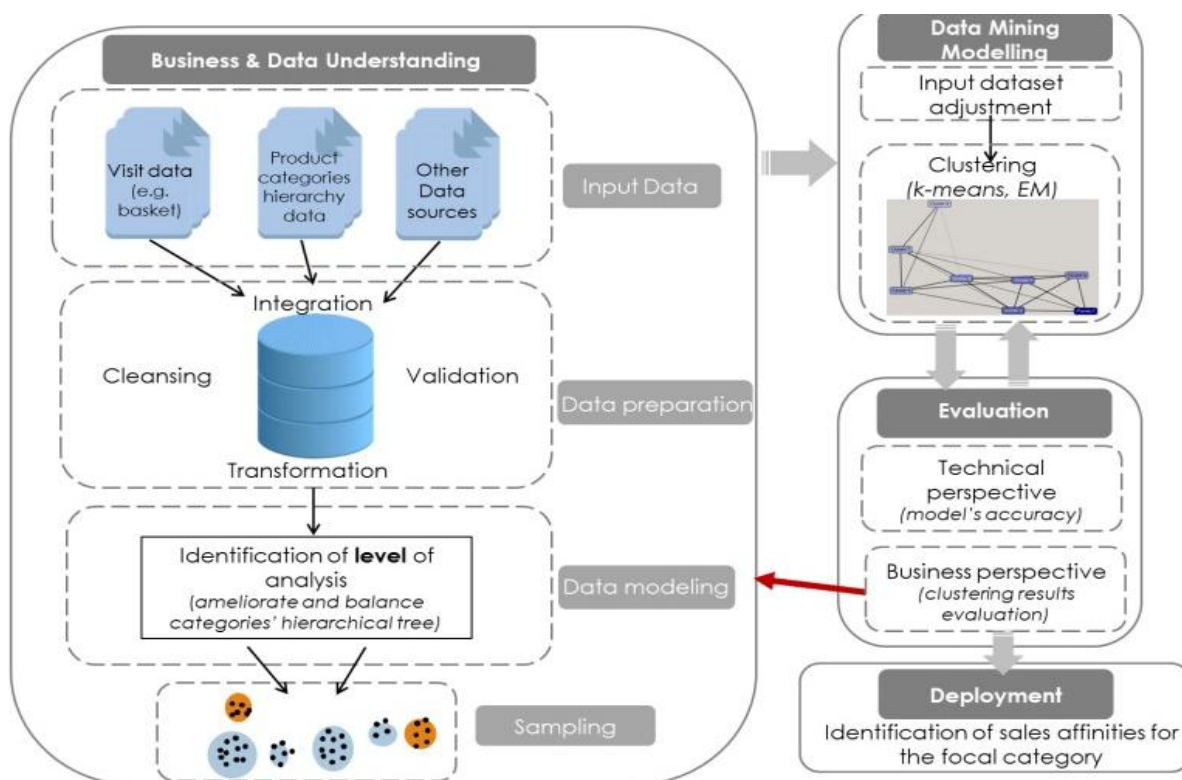


Fig 1: Generalized Data Mining Workflow for Customer Behavior Analysis (This figure shows a typical data mining process. How the data is handled and managed in finance sectors properly but have severe insider attacks to prevent it uses cybersecurity threat detection)

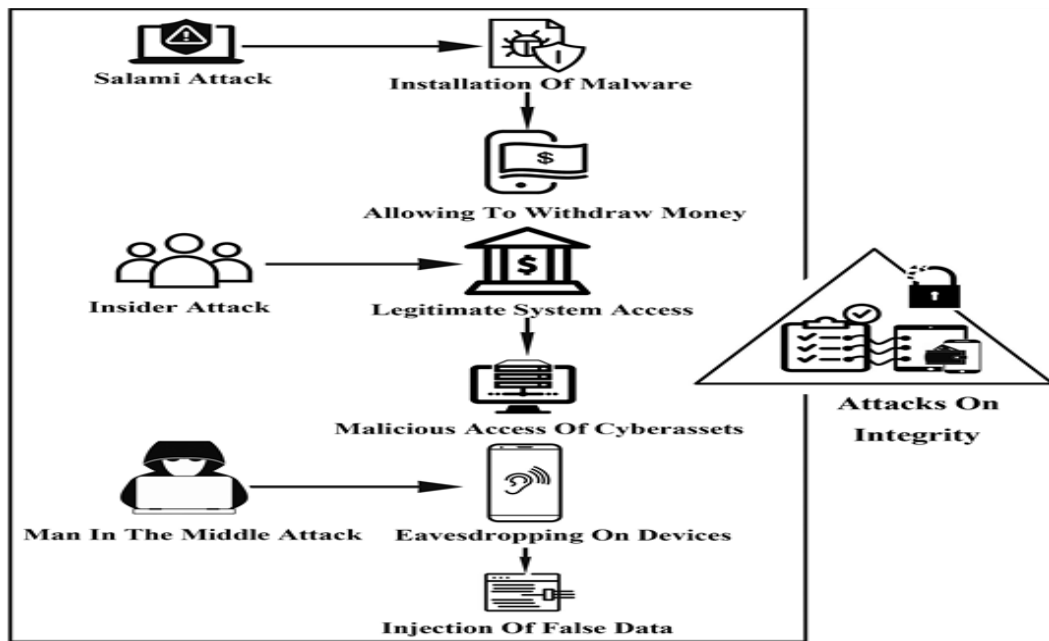


Fig 2: Cyber security Threat Detection and Action performed based on quick interruption analysis.

3 Technical Implementation

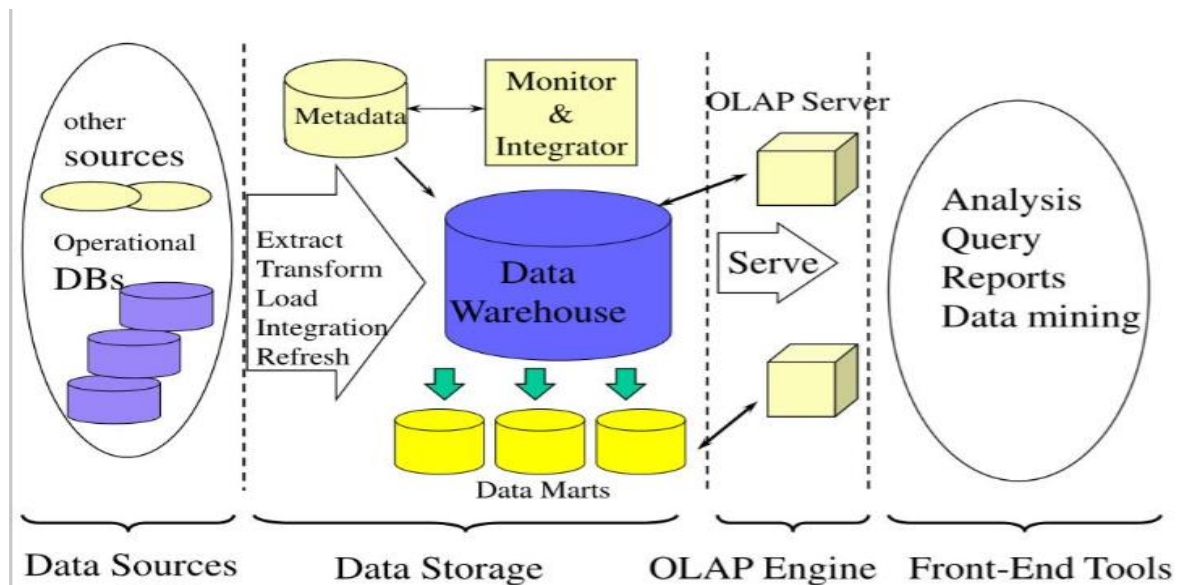


Fig 3: Multi-Tiered Data Mining Architecture for Behavioral Analysis and Insider Threat Detection in Enterprise and Financial Systems

It is a technology that aggregates structured data from one or more sources so that it can be compared and analyzed rather than transaction processing [7]. A data warehouse is designed to support the management decision-making process by providing a platform for data cleaning, data integration, and data consolidation [7],[8]. A data warehouse contains subject-oriented, integrated, time-variant, and non-volatile data. The Data

warehouse consolidates data from many sources while ensuring data quality, consistency, and accuracy. Data warehouse improves system performance by separating analytics processing from transactional databases. Data flows into a data warehouse from the various databases. A data warehouse works by organizing data into a schema that describes the layout and type of data. Query tools analyze the data tables using schema.

Bank and financial system data mining architecture begins with the data source layer that extracts vast volumes of structured as well as unstructured data from transactional systems, operational databases, and external sources like ATM logs, online banking portals, and payment gateways [9],[14]. Employee internal actions as well as customer actions are both present in such data. The **ETL (Extract, Transform, Load)** layer would then clean, normalize, and consolidate this raw data into a standardized format to maintain consistency and ready it for analysis, conforming to the norms of financial data handling norms [7],[9]. The processed data would then be routed to the data storage layer, where a central data warehouse holds the historical as well as real-time data securely. Within the warehouse, data marts are reserved for specific departments like audit, fraud identification, or compliance so that they can be easily accessed and analyzed. **OLAP (Online Analytical Processing)** engine plays a key role in making multidimensional analysis of behavioral and financial information possible to detect anomalies, patterns of fraud, or insider threat indicators using transaction frequency, time, location of access, and user behavior. This engine feeds into the OLAP server, which manages query processing and user access, such that financial data is safely and effectively served to authorized personnel [9].

Front-end tools at the topmost layer are where analysts, auditors, and decision-makers perform queries, generate reports, graph trends, and execute machine learning-based data mining algorithms such as clustering, classification, and anomaly detection to uncover threats or anomalies [7],[10]. Finally, the metadata and monitoring layer ensures end-to-end governance by storing data lineage information, processing user access logs, and continuously monitoring data flow for anomalies and malicious manipulation [6],[11]. This all-encompassing architecture not only facilitates secure, real-time management of financial data but also includes cyber forensic features for the effective detection, prevention, and diagnosis of insider threats [13].

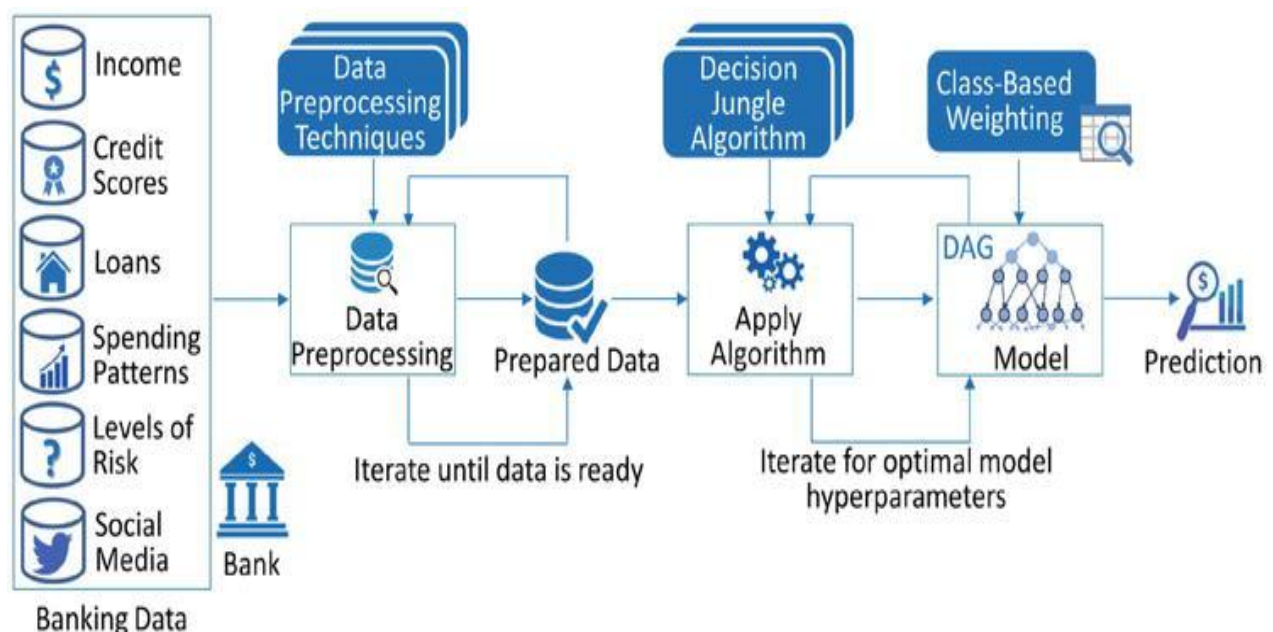


Fig 4: Working Structure of Machine Learning Pipelining for Risk Prediction and Threat Detection in Banking and Financial Systems in Data Warehousing and Data Mining

For the banking and finance industry, various types of information are collected, including income details, credit details, loan history, purchasing behavior, risk scores, and even social media information [5],[9]. Preprocessing is done in the steps of iteration until the data is structured completely and ready to use, and yield a firm Prepared Data set. The information is then inputted into the Machine Learning Stage, starting with the Decision Jungle Algorithm—a highly advanced development of decision trees that can process effectively on large-scale, sparse data [9]. The algorithm is used to identify patterns, correlations, and patterns in the past data, widely usable for

predicting customer behavior, default loans, or identifying insider threats based on behavioral irregularities. The entire technical stack is designed to be **robust, scalable, and compliant** with modern cybersecurity standards. Regular performance testing is conducted to measure system responsiveness, threat detection accuracy, and alert latency under simulated insider attack scenarios. Additionally, the architecture supports seamless integration with **Security Information and Event Management (SIEM)** systems and regulatory compliance modules, making it adaptable for deployment across a range of financial institutions. This modular and forward-compatible design ensures that the system not only addresses current insider threats but also evolves with emerging attack patterns and enterprise infrastructure needs.

3.1 Mathematical/Algorithmic Overview of Decision Jungle

- *Preprocess Data*

- *Initialize DAGs ensemble*

- *For each DAG:*

- *Split nodes by maximizing impurity reduction*
- *Allow node sharing to form DAG instead of tree*

- *Iterate to optimize hyperparameters*

- *Apply class-based weighting to handle imbalance*

- *Aggregate DAG predictions for final output*

The Decision Jungle algorithm builds an ensemble of **directed acyclic graphs (DAGs)** where nodes represent decision splits. Unlike traditional decision trees, DAGs allow shared nodes, reducing model complexity [8]. Training maximizes impurity reduction (**Gini or entropy**) [7]. Class-based weighting adjusts predictions to handle class imbalance. The final prediction aggregates **weighted outputs** from all DAGs. The algorithm also supports **feature importance extraction**, enabling analysts to interpret which behavioral or transactional features contribute most significantly to risk classification. This transparency is particularly valuable in **regulatory environments**, where explainability of AI decisions is a legal and ethical requirement. When applied to insider threat detection, Decision Jungle effectively distinguishes between normal and anomalous user behavior by capturing subtle sequential dependencies and behavioral deviations. Its resilience against noise and flexibility in handling high-dimensional, imbalanced datasets makes it an ideal choice for deployment in complex enterprise ecosystems.

Usage of blockchain technology in financial and business systems has significantly changed data security, especially when handling sensitive processes such as KYC (Know Your Customer) verification [5],[6]. As can be seen from the diagram, blockchain offers a decentralized and tamper-evident infrastructure wherein customer KYC information is loaded, verified, and shared by multiple banking departments and third-party companies securely and publicly [5]. This allows unproblematic intra-bank KYC and inter-bank communication with easily accessible confirmed customer data while providing for maximum levels of trust and data integrity [5]. As a means of guarantee that KYC updates are traceable and cannot be tampered with or manipulated illegally, blockchain makes it possible for tampering or illegal alteration by in-house stakeholders to be impossible [6].

In the case of insider behavioral risk, blockchain is an effective insider deterrent against insider behavior such as **data tampering, misuse of customer information, KYC scams**, insider leakage, and insider collusion to transfer fraudulent accounts [5],[6]. Since each transaction or change within the blockchain can be traced and read across the network by making use of cryptographic signatures, selfish insiders cannot erase or alter records. Besides this, smart contracts can automatically perform access controls and verifications with minimal human intervention and abuse of power [5]. Blockchain, combined with behavioral data mining and cyber forensic techniques, not only allows organizations to detect but also to proactively prevent insider threats, thereby having a secure, transparent, and accountable financial operations platform [1],[5].

3.2 Behavioral Data Protection in Banking and Enterprise Systems

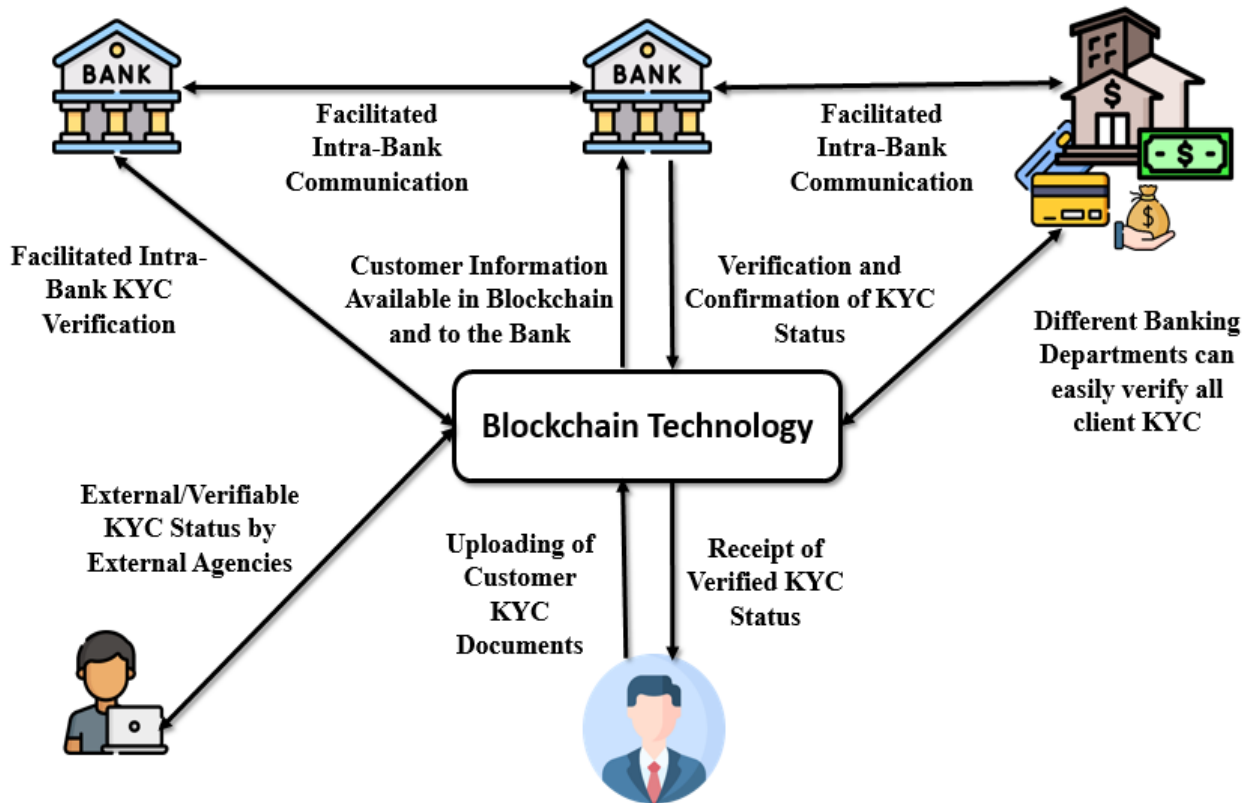


Fig 5: Cyber Forensic Framework for Protecting Sensitive Data in Financial and Enterprise Networks using Perspective using Block chain Technology KYC

In the case of insider behavioral risk, blockchain is an effective insider deterrent against insider behavior such as **data tampering, misuse of customer information, KYC scams**, insider leakage, and insider collusion to transfer fraudulent accounts [5],[6]. Since each transaction or change within the blockchain can be traced and read across the network by making use of cryptographic signatures, selfish insiders cannot erase or alter records. Besides this, smart contracts can automatically perform access controls and verifications with minimal human intervention and abuse of power [5]. Blockchain, combined with behavioral data mining and cyber forensic techniques, not only allows organizations to detect but also to proactively prevent insider threats, thereby having a secure, transparent, and accountable financial operations platform [1],[5].

Moreover, blockchain's **immutable ledger** enhances forensic investigation by providing a verifiable and chronological history of every user interaction, access event, and transaction attempt. This greatly simplifies the **audit trail process** and strengthens legal proceedings where proving intent, timeline, and authenticity of evidence is critical. In highly regulated financial environments, integrating blockchain with traditional **access control systems and threat intelligence platforms** ensures compliance with industry standards such as PCI-DSS, SOX, and GDPR. It also enables **cross-organizational coordination**, where verified behavioral logs can be securely shared between departments or partner institutions for collaborative risk mitigation, without compromising data privacy. By decentralizing control and ensuring trustless validation, blockchain creates a **tamper-resistant infrastructure** that not only deters insider actions but also fosters a culture of accountability and operational transparency across all levels of the organization.

3.3 Behavioral Insider Attacks which may cause fraudulent Transactions

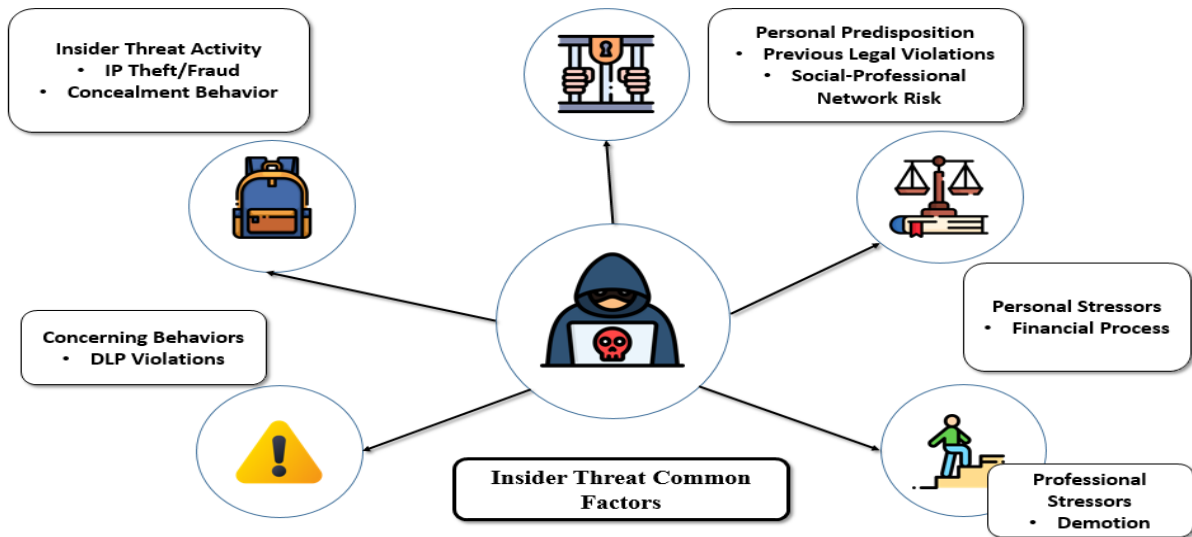


Fig 5: Cyber Forensic Framework for Protecting Sensitive Data in Financial and Enterprise Networks

The image highlights key behavioral factors contributing to insider threats in enterprise systems. These include **personal predispositions** (like past legal issues or risky associations), **personal stressors** (such as financial or legal problems), and **professional stressors** [2],[13]. Insider threat activities may involve **IP theft, fraud, or concealment behaviors**, often accompanied by **DLP violations or code of conduct breaches** [4],[13].

4 Evaluation Framework and Data Workflow in Cybersecurity for Secure Data Transactions

The framework essentially incorporates a number of key elements. First, User Authentication and Access Control measures such as **multi-factor authentication (MFA) and role-based access control (RBAC)** guarantee that only legitimate users are allowed to execute transactions. Second, Data Encryption safeguards all transactional data, in transit as well as at rest, with industry norms like **AES-256** and secure communication protocols like **TLS 1.3 or SSL** [11]. Third, Tokenization and Masking strategies safeguard sensitive data

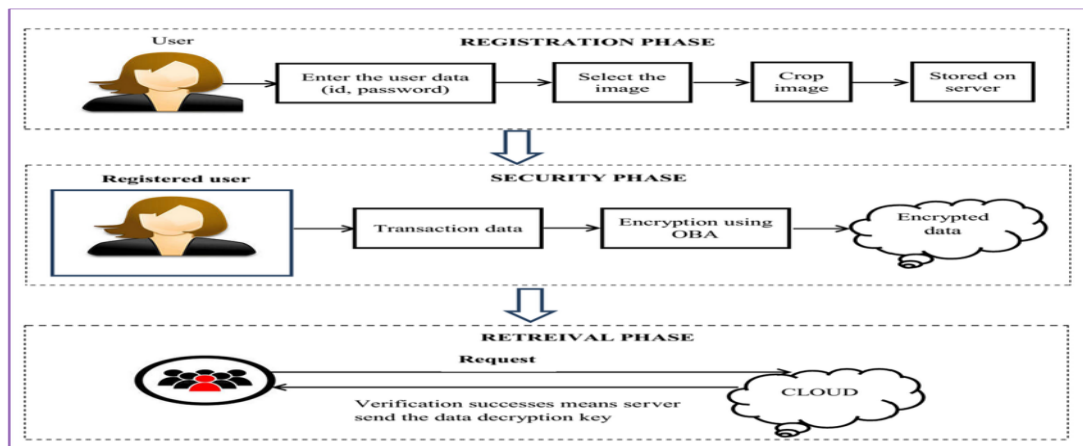


Fig 6: Cryptographic Techniques used for secure transactions using each User's Private ID

4.1 Data Workflow & Management via Data Mining

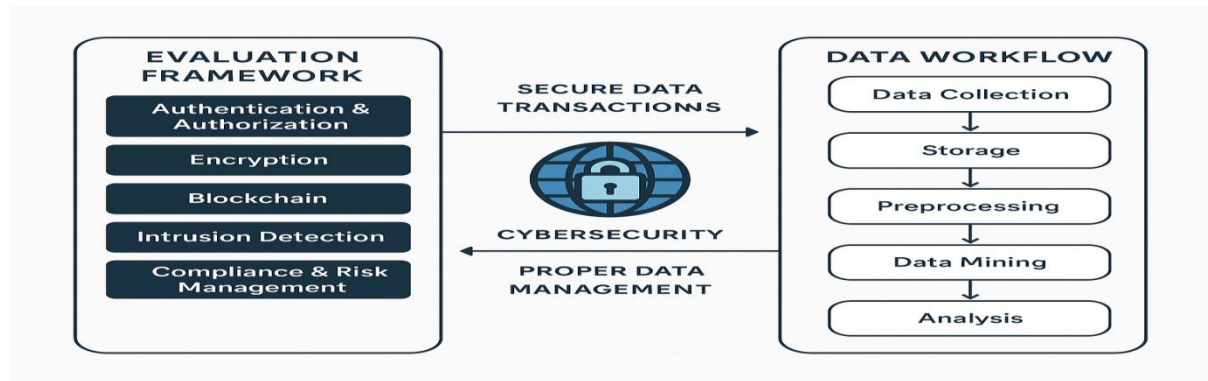


Fig 7: Data Workflow in Cyber security for Secure Transactions

5 Future Scope and Directions

In the near future, ‘data mining’ and ‘cybersecurity’ will continue to develop and have a tremendous impact on the processing, management, and securing of data across sectors [8],[11]. Data mining will get even smarter and contextual, giving real-time insight, anomaly identification, and predictive analysis with greater accuracy. It will assist companies not only in handling large amounts of data but also in revealing hidden patterns that can aid strategic decisions, fraud detection, and user-centric experiences [7],[8],[12].

In the future, insider threat detection systems will increasingly incorporate **context-aware AI models** that can adapt to user roles, time-based patterns, and system sensitivity levels, allowing for finer-grained detection of abnormal behaviour.

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